Shape of the distribution

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Shape of the distribution

- In previous chapter, we learned about “Location” and “Dispersion” which are two important quantitative concepts.

- Two data sets may have identical means and identical variance, but their graphical shapes may be different

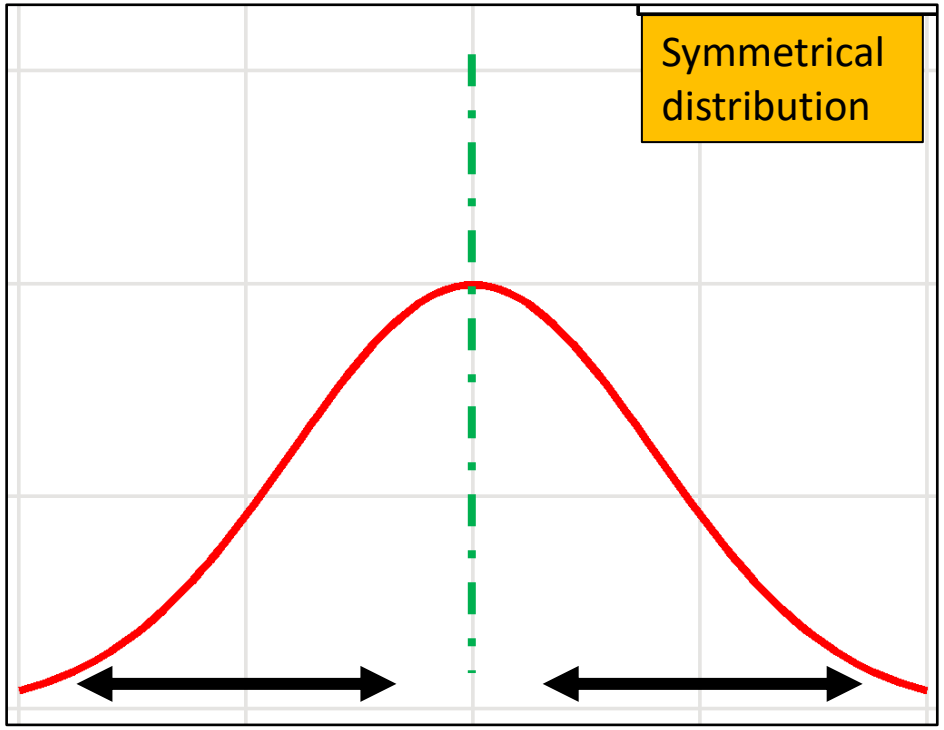
Shape of the distribution

Identified by using two characteristics :

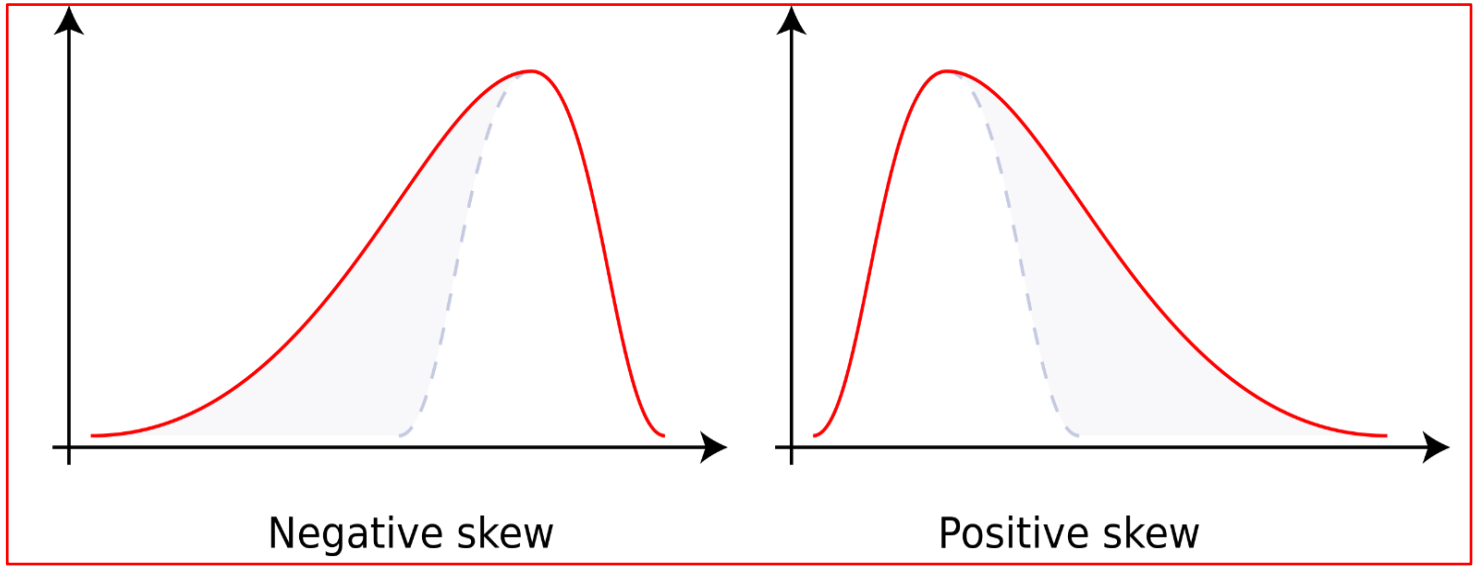
1. Skewness
2. Kurtosis

Skewness

- Lack of symmetry of a distribution



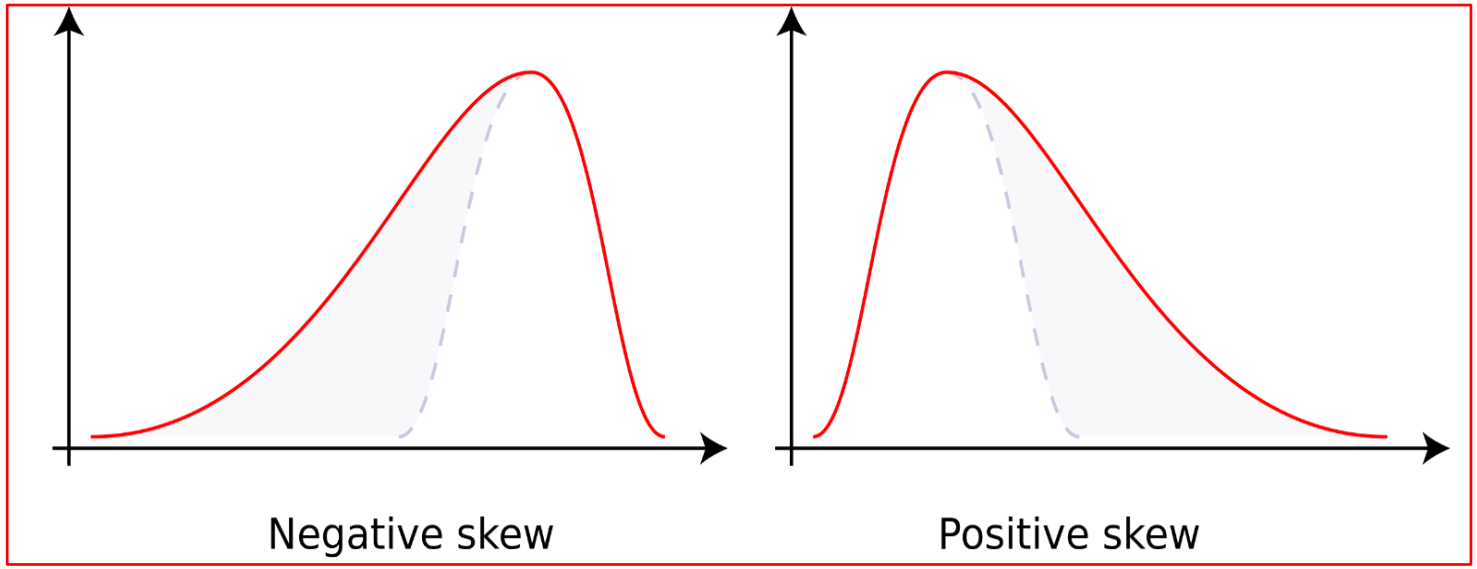
Lack of symmetry



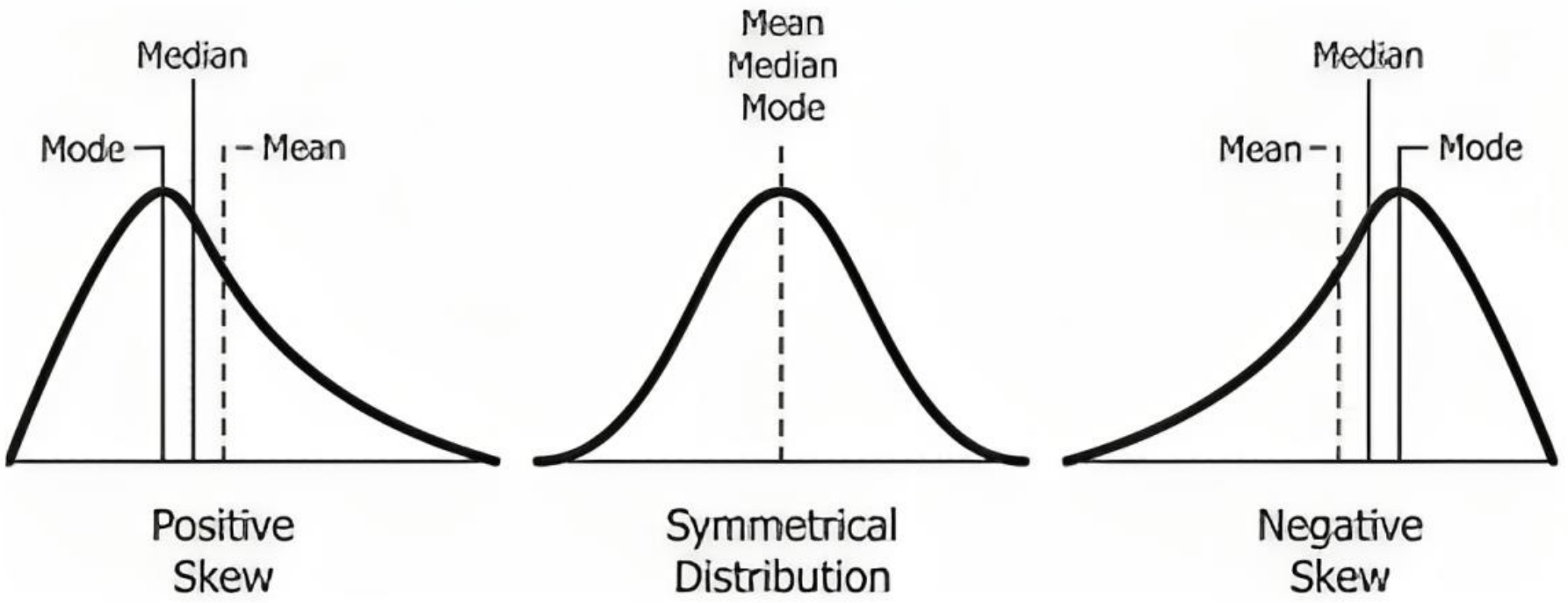
Skewness

- There are two types of skewness or lack of symmetry occurs:

1. Positive skewness
2. Negative skewness



Graph



$Mean > Median > Mode$

$Mean < Median < Mode$

Coefficient of skewness

- $S_k = \frac{3(Mean - Median)}{Standard\ Deviation}$

- $S_k > 0$: ***Positively skewed***
- $S_k < 0$: ***Negatively skewed***
- $S_k = 0$: ***Symmetric***

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For example: Calculate the coefficient of skewness: 9,8,5,14,16

Mean = 10.4
Median = 9
SD = 4.5

$$S_k = \frac{3(Mean - Median)}{SD} = \frac{3(10.4 - 9)}{4.5}$$
$$= 0.93$$

Since $S_k > 0$, thus the distribution is positively skewed distribution.

Kurtosis

- $If\ r = 1, \mu_1 = First\ moment = \frac{\sum(X_i - \bar{X})^1}{N}$
- $If\ r = 2, \mu_2 = Second\ moment = \frac{\sum(X_i - \bar{X})^2}{N}$
- $If\ r = 3, \mu_3 = Third\ moment = \frac{\sum(X_i - \bar{X})^3}{N}$
- $If\ r = 4, \mu_4 = Fourth\ moment = \frac{\sum(X_i - \bar{X})^4}{N}$

- Degree of peaked or flatness of a distribution

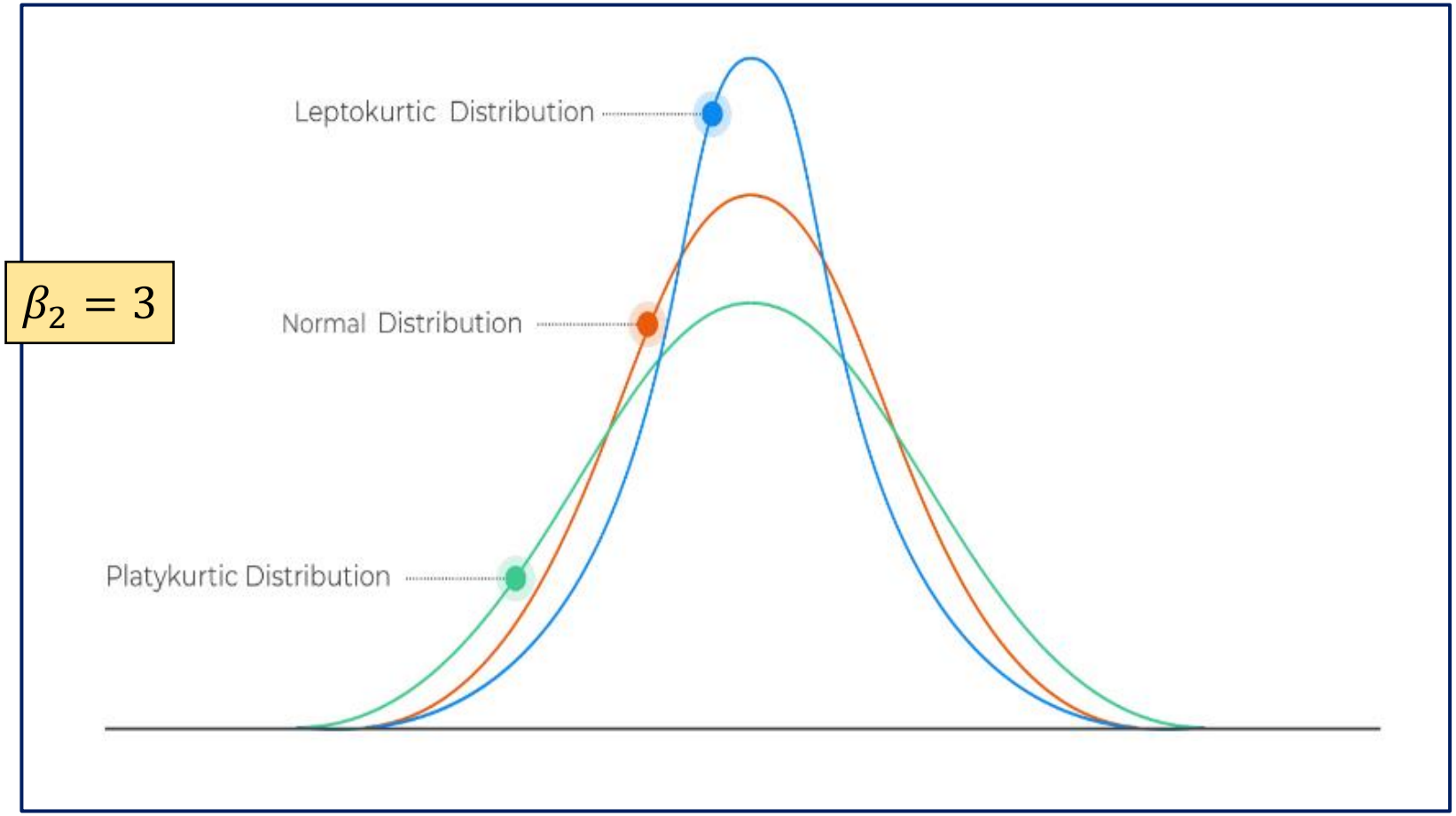
- Three types-

a) Leptokurtic $\beta_2 > 3$

b) Mesokurtic/Normal $\beta_2 = 3$

c) Platykurtic $\beta_2 < 3$

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$



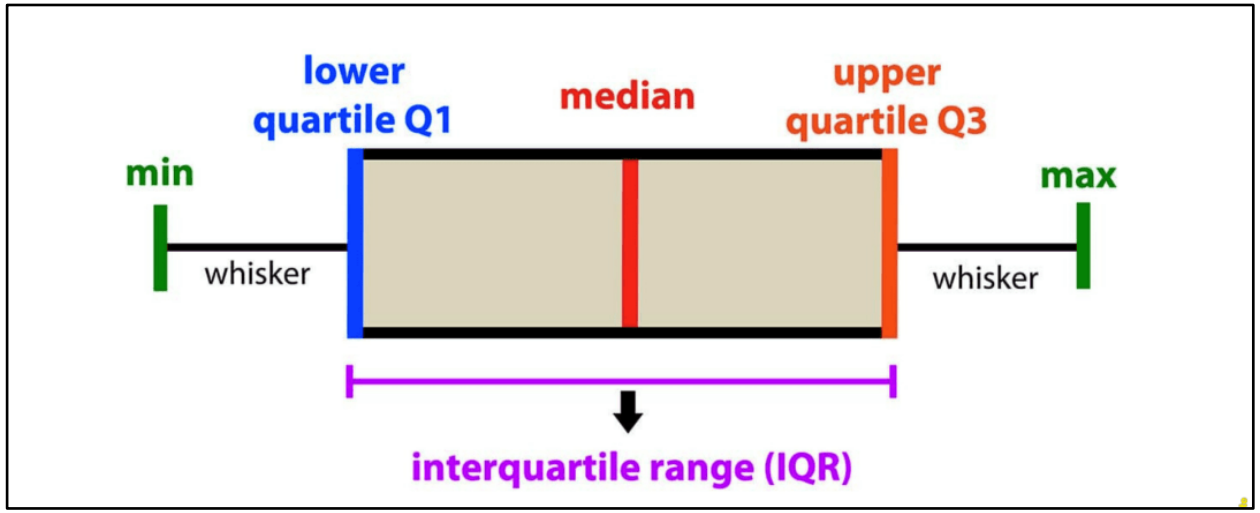
Box & Whisker plot

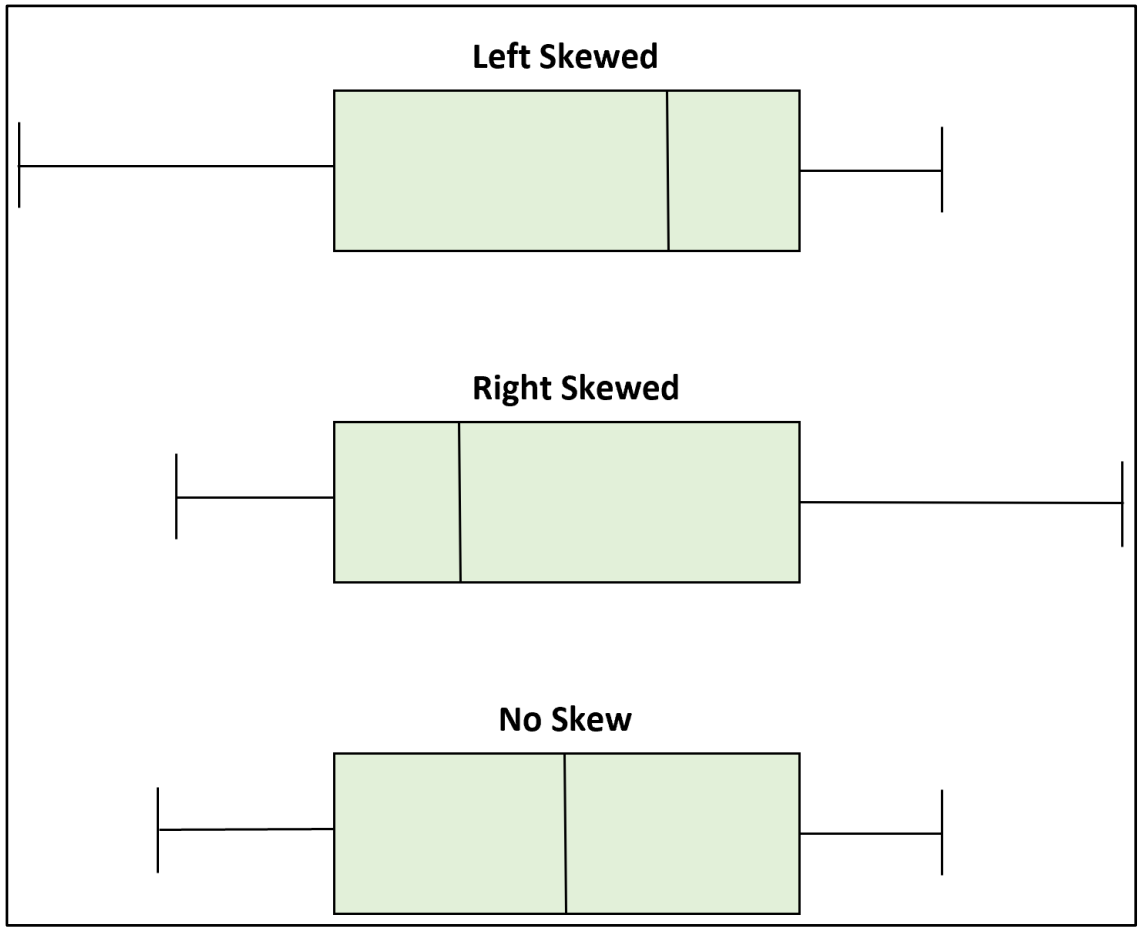
- A box-whisker plot, also known as a box plot, is a graphical representation of the distribution of a dataset.
- It displays the five-number summary of the data:

1. Minimum value
2. First Quartile (Q1)
3. Second Quartile/Median (Q2)
4. Third Quartile (Q3)
5. Maximum value

Importance of Box plot

- To get idea of the shape of the distribution
- To detect outliers from the data.



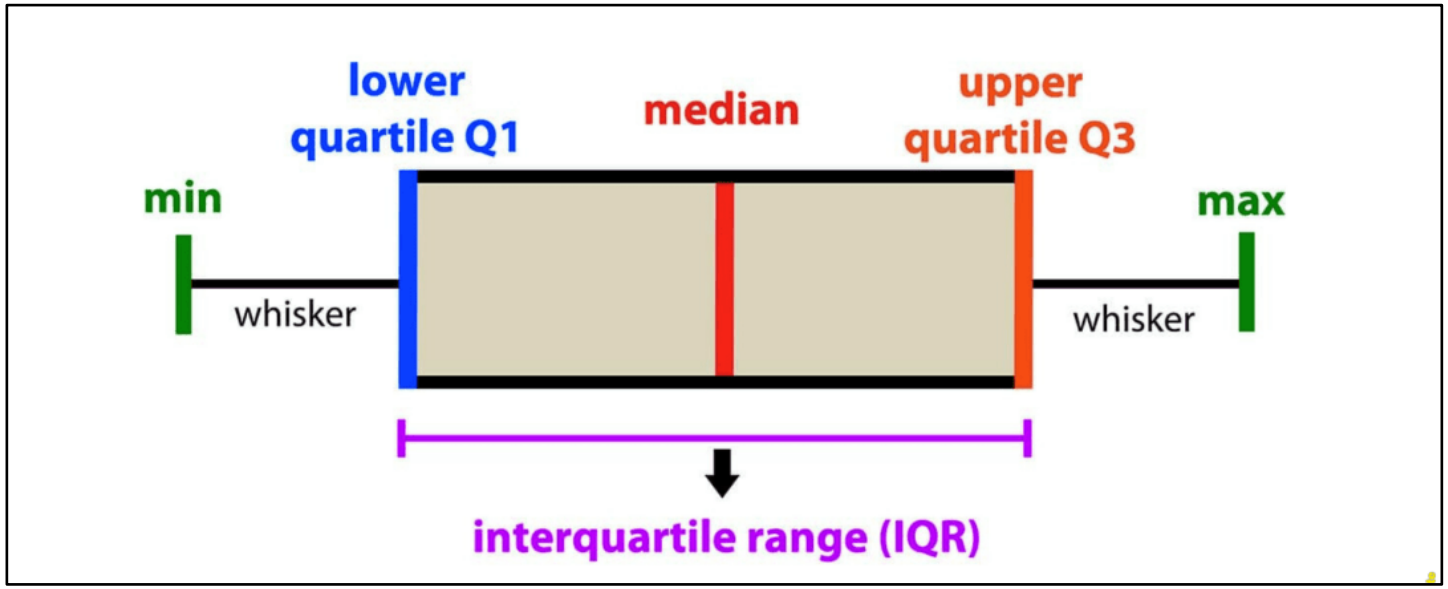


Identify outliers from boxplot

$$IQR = Inter\ Quartile\ Range$$
$$IQR = Q_3 - Q_1$$

Lower Fences:
$$\{Q_1 - 1.5 \times IQR\}$$

Upper Fences:
$$\{Q_3 + 1.5 \times IQR\}$$



Identify outliers from boxplot

512,643,784,275,398,529,610,475,382,491,566,734

Draw the box-whisker plot and detect the outlier(s)

Organize the data into ascending order,

275,382,398,475,491,512,529,566,610,643,

734,784

Now,

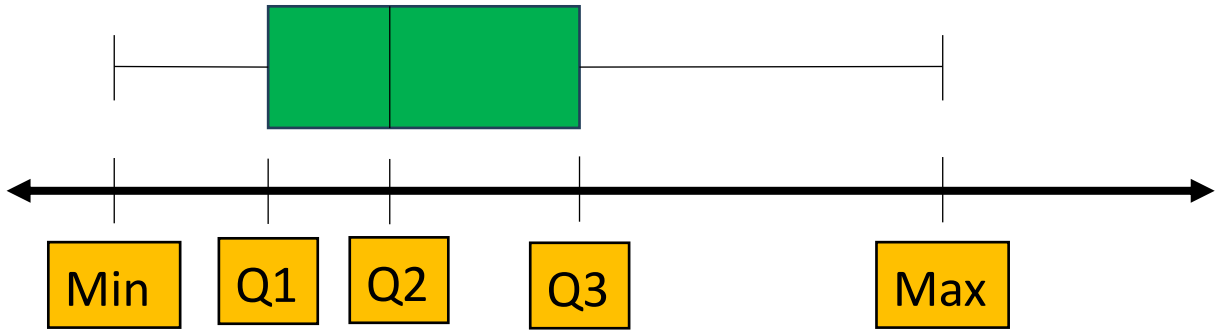
Min=151.5

Q1=436.5

Q2=520.5

Q3=626.5

Max=911.5



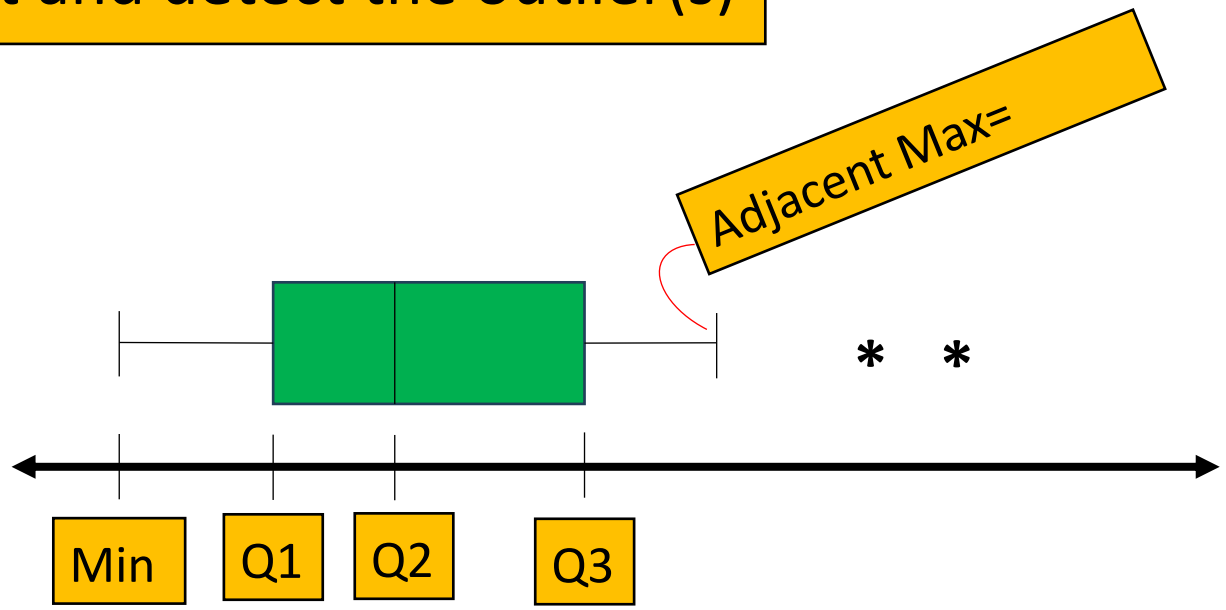
Identify outliers from boxplot

Draw the box-whisker plot and detect the outlier(s)

$$\therefore IQR = Q_3 - Q_1 = 190$$

$$Lower = \{Q_1 - 1.5 \times IQR\} =$$

$$Upper = \{Q_3 + 1.5 \times IQR\} =$$



Thank You!